Discrete mathematics

1. Language

Definition 1.1 (Semantics)

The meaning of a sequence of symbols.

Definition 1.2 (Syntax)

The grammatical rules for composing symbols into valid sentences.

Definition 1.3 (Variable)

A symbol that stands in place for an object that has not been determined yet.

A name can be assigned to a particular object using $:=$.

Definition 1.4 (Sentence)

A sentence is the expression of an idea in accordance of the syntax and grammar of a given language.

A statment is ***atomic*** if it cannot be broken down into smaller components while complying with the syntax and grammar.

- **Declarative:** Describes something. Usually, it describes a ***subject*** and a property, called ***predicate***, that the subject has.
- **Interrogative:** Asks a non-rhetorical question.
- **Imperative:** Command or request.

In math, we are concerned with declarative sentences about mathematical objects.

1.1. Church-Turing thesis

Definition 1.1.1 (Computable)

A function is ***computable*** if any of the following equivalent things are true:
- It is expressible as a general recursive process.
- It is a term in the λ-calculus.
- It can be described by a Turing machine.

Axiom 1.1.2 (Church-Turing Thesis)

Our intuitive definition of "computable" (which Turing calls "effectively calculable") is equivalent to the above definition.

2. Zeroth-order logic

2.1. Truth values

See Cedre DM1.1

Definition 2.1.1 (Truth value)

There are only two truth values:
- $\top$ "true"
- $\bot$ "false"

Definition 2.1.2 (Propositional equivalence)

Two propositions p and q are logically equivalent, denoted $p \equiv q$, iff they have the same truth value.

Definition 2.1.3

R is ***reflexive*** iff aRa for all $a \in A$.

R is ***symmetric*** iff $aRb \Leftrightarrow bRa$.

R is ***transitive*** iff $aRb, bRc \Rightarrow aRc$.

Definition 2.1.4 (Equivalence relation)

A relation that is reflexive, symmetric, and transitive.

Theorem 2.1.5

Logical equivalence is an equivalence relation.

Axiom 2.1.6 (Principle of Bivalence)

Sentences expressing truth values are either true or false, but not both.

Definition 2.1.7 (Proposition (informal definition))

A statement which is either true or false, i.e. a statement expressing a truth value.

2.2. Logical connectives

See Fleck 2.2~2.9

See Cedre DM1.2, CL00

Definition 2.2.1 (Negation)

$$\neg\top = \bot$$
$$\neg\bot = \top$$

Definition 2.2.2 (Logical dual)

Logical connectives f and g are logical duals iff $\neg f = g$ and $\neg g = f$ on all possible inputs.

Definition 2.2.3 (Conjunction, disjunction)

- **Conjunction:** $p \land q = p \text{ AND } q$
- **Disjunction:** $p \lor q = p \text{ OR } q$ (inclusive or)

Conjunction and disjunction are dual.

Definition 2.2.4 (Conditional)

$p \to q$ means "if p, then q". p is the ***antecedent*** and q is the ***consequent***.

If p and q are propositional formulae: $p \to q$ t is true iff every time p is satisfied, then q is satisfied.

If p and q are propositions: it is false iff p is true and q is false.

Warning: If p is always false, then $p \to q$ is always true regardless if q is false! Also, there need not be any causal relationship between p and q.

Definition 2.2.5 (Converse)

The ***converse*** of $p \to q$ is $q \to p$. These are not equivalent.

Definition 2.2.6 (Contrapositive)

The ***contrapositive*** of $p \to q$ is $\neg q \to \neg p$. It is equivalent to the original statement:

$$p \to q \equiv \neg q \to \neg p$$

Definition 2.2.7 (Biconditional)

$$p \longleftrightarrow q = p \text{ iff } q$$
$$p \longleftrightarrow q \implies (p \to q) \land (q \to p)$$

Definition 2.2.8 (Exclusive disjunction)

$$p \oplus q = p \text{ XOR } q = p \longleftrightarrow .\neg q$$

p	q	$\neg p$	$p \land q$	$p \lor q$	$p \oplus q$	$p \to q$	$p \longleftrightarrow q$
$\top$	$\top$	$\bot$	$\top$	$\top$	$\bot$	$\top$	$\top$
$\top$	$\bot$	$\bot$	$\bot$	$\top$	$\top$	$\bot$	$\bot$
$\bot$	$\top$	$\top$	$\bot$	$\top$	$\top$	$\top$	$\bot$
$\bot$	$\bot$	$\top$	$\bot$	$\bot$	$\bot$	$\top$	$\top$

Definition 2.2.9 (Proposition)

λ is a proposition iff λ satisfies any of the following conditions:

- $\lambda = \top$ or $\lambda = \bot$.
- $\lambda = \neg\varphi$, where φ is a proposition.
- $\lambda = \varphi \land \psi$, where φ and ψ are propositions.
- $\lambda = \varphi \lor \psi$, where φ and ψ are propositions.
- $\lambda = \varphi \to \psi$, where φ and ψ are propositions.
- $\lambda = \varphi \longleftrightarrow \psi$, where φ and ψ are propositions.

We use this definition inductively, by first establishing that $\top$ and $\bot$ are propositions, and then using the connectives to compose larger and larger propositions. Alternatively, we use this definition recursively to determine that statements are propositions by recursively decomposing them.

2.3. Propositional formulae

See Cedre DM1.2

Definition 2.3.1 (Propositional formula)

An expression that evaluates as a proposition when all of its variables are themselves replaced by propositions.

Definition 2.3.2 (Logical equivalence and nonequivalence)

Let φ and ψ be propositional formulae both consisting of the same variables $p_1, ...p_n$. $\varphi \equiv \psi$ iff every assignment of truth values to those variables produces the same truth value.

Theorem 2.3.3

Computing a truth table to check if two propositional formulae of n variables are equivalent requires 2^n rows.

Thus, we want to prove that the two expressions are equivalent, rather than brute-forcing them.

2.4. Propositional logic

See Cedre DM1.3 *See* Fleck 2.2~2.9

Axiom 2.4.1 (Axioms of classical logic)

The following are the axioms of ***Boolean algebra***, and they have two equivalent forms:

Identity	$\top \land p \equiv p$	$\bot \lor p \equiv p$
Complement	$\neg p \land p \equiv \bot$	$\neg p \lor p \equiv \top$
Commutativity	$p \land q \equiv q \land p$	$p \lor q \equiv q \lor p$
Associativity	$p \land (q \land r) \equiv (p \land q) \land r$	$p \lor (q \lor r) \equiv (p \lor q) \lor r$
Distributivity	$p \land (q \lor r) \equiv (p \land q) \lor (p \land r)$	

We also have the following two axioms:

- **Conditional disintegration:** $p \to q \equiv \neg p \lor q$
- **Biconditional disintegration:** $p \longleftrightarrow \equiv (p \to q) \land (q \to p)$

Theorem 2.4.2 (Uniqueness of complements)

For any p and q, if $p \land q \equiv \bot$ and $p \lor q \equiv \top$, then $\neg p \equiv q$.

Corollary 2.4.2.1

- $\top \equiv \neg\bot$ and $\bot \equiv \neg\top$
- $p \equiv q \iff \neg p \equiv \neg q$
- For any p, q, r, s s.t. $p \equiv q$ and $r \equiv s$:
  - $p \land r \equiv q \land s$
  - $p \lor r \equiv q \lor s$
  - $p \to r \equiv q \to s$
  - $p \longleftrightarrow r \equiv q \longleftrightarrow s$

Theorem 2.4.3 (Idempotence)

For any proposition p, $p \land p \equiv p$ and $p \lor p \equiv p$.

Theorem 2.4.4 (Domination)

For any proposition p, $\top \lor p \equiv \top$ and $\bot \land p \equiv \bot$.

Theorem 2.4.5 (De Morgan's Laws)

- $\neg(p \land q) \equiv \neg p \lor \neg q$
- $\neg(p \lor q) \equiv \neg p \land \neg q$

Theorem 2.4.6 (Negation equivalences)

- $\neg(\neg p) \equiv p$

- $\neg(p \to q) \equiv p \land \neg q$

2.5. Rules of inference

Definition 2.5.1 (Turnstile symbol)

$\vdash \varphi$, where φ is a predicate, means that we can prove φ with what we currently know.

$\Gamma \vdash \varphi$ means that given that assumptions Γ are true, we can prove that φ is true.

Axiom 2.5.2 (Deduction rule)

If by assuming p, we can prove q, then we can write $p \to q$:

$$(p \vdash q) \vdash (p \to q).$$

Axiom 2.5.3 (Modus ponens)

If we have $p \to q$ and we know p is true, then we can deduce q is true:

$$p, (p \to q) \vdash q.$$

Axiom 2.5.4 (Reductio ab absurdum)

If $\neg p$ leads to a contradiction, then $\neg p$ is absurd; thus p is true:

$$(\neg p \vdash q), (\neg p \vdash \neg q) \vdash p.$$

Theorem 2.5.5 (Modus tollens)

If we have $p \to q$ but also $\neg q$, then we can infer $\neg p$:

$$\neg q, (p \to q) \vdash \neg p.$$

2.5.1. Syllogisms and other theorems

Hypothetical syllogism:

$$(p \to q), (q \to r) \vdash p \to r$$

Implication elimination: (consolidation rule)

$$(p \to q) \vdash (p \vdash q)$$

Conjunction introduction: (adjunction)

$$p, q \vdash p \land q$$

Conjunction elimination: (simplification)

$$p \land q \vdash p$$

Disjunction introduction: (proof by cases)

$$p \vdash p \lor q$$

Disjunction elimination: (explosion)

$$(p \to r), (q \to r), (p \lor q) \vdash r$$

Constructive dilemma:

$$(\alpha \to \gamma), (\beta \to \delta), (\alpha \lor \beta) \vdash \gamma \lor \delta$$

3. First-order logic

See Cedre DM2, Cedre CL00, Cedre CL01

3.1. Objects

Definition 3.1.1 (Universe of discourse)

The collection of all objects that we "care about" / we "discuss".

Definition 3.1.2 (Term)

A symbol that refers to an object in the universe of discourse.

A term which refers to a specific object is a ***constant***.

Definition 3.1.3 (Variable)

A ***variable*** is a term that refers to a generic object, not one particular object.

In order for a variable to be meaningful, it must be quantified (for example by a quantifier like $\forall$ or $\exists$) within some scope, i.e. a ***bound variable***. A variable which is not quantified is a ***free variable***.

3.2. Predicates

Definition 3.2.1 (Predicate)

Let $x_1, ..., x_n$ be variables. $\varphi(x_1, ..., x_n)$ is an ***n-ary predicate*** iff replacing each of the n variables by constants $t_1, ..., t_n$ results in a proposition $\varphi(t_1, ..., t_n)$ carrying a truth value.

3.3. Quantification

Definition 3.3.1 (Universal quantification)

The ***universal quantification*** of x appearing in φ is denoted

$$\forall x(\varphi(x, y_1, ..., y_n)).$$

It means that any constant replacing x will satisfy φ.

Definition 3.3.2 (Existential quantification)

The ***existential quantification*** of x appearing in φ is denoted

$$\exists x(\varphi(x, y_1, ..., y_n)).$$

It means that there exists some value of x which satisfies φ.

Definition 3.3.3 (Unique existential quantification)

There exists a unique x s.t. $\varphi(x)$.

$$\exists! x(\varphi(x)) \coloneqq \exists x(\varphi(x) \land y(\varphi(y) \to (y = x))).$$

3.4. Formulae

Definition 3.4.1 (Atomic formula)

A formula φ is ***atomic*** iff it satisfies one of the following:

- $\varphi = \top$ or $\varphi = \bot$.
- $\varphi = \psi(t_1, ..., t_n)$ where ψ is an n-ary predicate and $t_1, ..., t_n$ are terms.

Definition 3.4.2 (Well-formed formula)

A formula λ is ***well-formed*** (is a ***wff***) iff it satisfies one of the following:

- λ is an atomic formula.
- $\lambda = \neg\varphi$, where φ is a wff.
- $\lambda = \varphi \land \psi$, where φ and ψ are wff.
- $\lambda = \varphi \lor \psi$, where φ and ψ are wff.
- $\lambda = \varphi \to \psi$, where φ and ψ are wff.
- $\lambda = \varphi \longleftrightarrow \psi$, where φ and ψ are wff.
- $\lambda = \forall x(\varphi)$, where φ is a wff.
- $\lambda = \exists x(\varphi)$, where φ is a wff.

A well-formed formula with no free variables is called a ***sentence in the first-order logic***.

3.5. Rules of inference

Axiom 3.5.1 (Universal introduction)

If $\varphi(t)$ for an arbitrary term t, then $\forall x(\varphi(x))$.

Let t be arbitrary. Then,

$$\varphi(t) \vdash \forall x(\varphi(x)).$$

Axiom 3.5.2 (Universal elimination)

If $\forall x(\varphi(x))$, then we can pick any t and $\varphi(t)$ is true.

Choose some t. Then,

$$\forall x(\varphi(x)) \vdash \varphi(t).$$

Axiom 3.5.3 (Existential introduction)

If we know $\varphi(t)$ for a specific term t, then $\exists x(\varphi(x))$.

Choose some t. Then,

$$\varphi(t) \vdash \exists x(\varphi(x)).$$

Axiom 3.5.4 (Existential elimination)

If we know $\exists x(\varphi(x))$, then $\varphi(t)$ for some new t that has not yet appeared.

$$\exists x(\varphi(x)) \vdash \varphi(t)$$

where t comes out of the assertion.

3.6. Useful theorems

Theorem 3.6.1 (Negation of quantifiers)

For any well-formed formula φ with one free variable,

$$\neg\forall x(\varphi(x)) \equiv \exists x(\neg\varphi(x))$$
$$\neg\exists x(\varphi(x)) \equiv \forall x(\neg\varphi(x)).$$

Theorem 3.6.2 (Quantifier shift)

For any well-formed formula φ containing two free variables,

$$\forall x\forall y(\varphi(x,y)) \equiv \forall y\forall x(\varphi(x,y))$$
$$\exists x\exists y(\varphi(x,y)) \equiv \exists y\exists x(\varphi(x,y)).$$

This means that we can commute quantifiers when they are the same; however the universal and existential quantifiers don't necessarily commute with each other:

$$\exists x\forall y(\varphi(x,y)) \vdash \forall y\exists x(\varphi(x,y))$$
$$\forall x\exists y(\varphi(x,y)) \nvdash \exists y\forall x(\varphi(x,y)).$$

Theorem 3.6.3 (Distribution of quantifiers)

For any well-formed formulae φ and ψ, each containing exactly one free variable, we can distribute quantifiers as shown below:

$$\forall x(\varphi(x) \land \psi(x)) \equiv \forall x(\varphi(x)) \land \forall x(\psi(x)).$$
$$\forall x(\varphi(x) \lor \psi(x)) \equiv \exists x(\varphi(x)) \lor \exists x(\psi(x)).$$

This means that universal quantifiers can be distributed over conjunctions, and existential quantifiers can be distributed over disjunctions. The opposite does not hold.